
Math Training — M1 Cognitive Science

Session 5 — Analysis

Functions, derivatives, integrals, several variables, one ODE

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Nothing beyond arithmetic is assumed. We build the tools from the ground up, in the order they depend on each other.

- **Functions:** what they are, and the shapes that keep coming back — especially the *exponential*.
- **Derivatives:** how fast something is changing, and what the *sign* tells you.
- **Integrals:** adding up change, and area under a curve.
- **Several variables:** functions with more than one input, and functions whose output is a *vector*.
- **Differential equations:** what they say, and how to solve the one that matters.

■ If you remember only one thing

A **derivative** says how fast something is changing right now. An **integral** adds those changes back up. A **differential equation** tells you the rate in terms of the thing itself — and to *solve* it is to say what the thing *does*.

That last sentence also gives you a free check on every answer you will ever write: differentiate it and see whether the equation comes back.

■ Definition

A **function** is a rule that turns one number into another: give it t , it gives back $f(t)$. Its **graph** is what you get by marking the height $f(t)$ above each t .

Two words worth keeping straight:

- the **variable** is what you feed in — time, difficulty, number of practice trials;
- the **parameters** are the numbers that decide *which* function you have, and they stay fixed as the variable runs.

In $R(t) = 0.9e^{-t/2}$, the variable is t . The numbers 0.9 and 2 are parameters.

Three shapes are almost all we need

- **Constant**, $f(t) = c$: a flat horizontal line. Nothing changes.
- **Linear**, $f(t) = mt + c$: a straight line climbing by m per step, with c its value at $t = 0$.
- **Exponential**, $f(t) = e^{at}$: the shape the whole session runs on.

Powers such as t^2 , $\sqrt{t} = t^{1/2}$ and $1/t = t^{-1}$ appear too, and the logarithm $\ln t$ shows up as the thing that undoes an exponential: $\ln(e^x) = x$.

■ Definition

$e \approx 2.718$ is the one base for which the curve e^t climbs, at every point, at exactly the height it has reached. What we need are its consequences:

$$e^0 = 1, \quad e^{at} > 0 \text{ always}, \quad e^{at} \text{ grows if } a > 0, \text{ decays if } a < 0.$$

For $a < 0$ write $a = -1/\tau$, so the function is $e^{-t/\tau}$ and the positive number τ is the **time constant**.

■ Key idea

After a time τ , the quantity has fallen to $e^{-1} \approx 37\%$ of its starting value — and after another τ to 37% of *that*, for ever. The curve approaches zero without ever arriving.

■ Link to cognitive science

Learn a list of words, then test recall after a delay t . The proportion still recalled falls off as

$$R(t) = R_0 e^{-t/\tau},$$

the shape Ebbinghaus measured on himself in 1885 and the standard first description of forgetting ever since. R_0 is what you recall immediately; τ is how long the memory lasts, in the sense on the previous slide.

With $R_0 = 0.9$ and $\tau = 2$ days:

$$R(0) = 0.9, \quad R(2) = 0.9 \times 0.37 \approx 0.33, \quad R(6) = 0.9 \times 0.05 \approx 0.05.$$

Two parameters, and the entire curve. Notice that all the arithmetic used the three memorised numbers and nothing else.

■ Practice — try it

1. Does e^{3t} grow or decay? And e^{-3t} ?
2. Write $e^{-t/5}$ in the form e^{at} and give a and τ .
3. Recall follows $R(t) = 0.8e^{-t/3}$, with t in days. Give $R(0)$, $R(3)$ and $R(9)$.
4. Two students forget with $\tau = 2$ days and $\tau = 8$ days. After 8 days, who remembers more, and roughly how much does each have left?

■ Definition

For a straight line, “how fast is it climbing” is the **slope**: rise divided by run. For a curve the slope changes from point to point, so we do the same thing *locally* — zoom in far enough and any smooth curve looks straight, and the slope of that line is the **derivative**, written $f'(t)$ or $\frac{df}{dt}$.

When t is time, the derivative is a **rate of change per unit time**. When the variable is something else — difficulty, dose, number of trials — it is still “how much does the output move when I nudge the input”.

■ Useful fact

$$\text{constant} \rightarrow 0, \quad t^n \rightarrow n t^{n-1}, \quad (\alpha f + \beta g)' = \alpha f' + \beta g', \quad \boxed{e^{at} \rightarrow a e^{at}}$$

The power rule holds for negative and fractional n too, so $1/t \rightarrow -1/t^2$ and $\sqrt{t} \rightarrow 1/(2\sqrt{t})$.

■ Intuition

The boxed rule in words: differentiating an exponential gives the *same* exponential back, multiplied by whatever sits in front of the t in the exponent. The exponent does not change and the base does not change.

That single fact is why exponentials — and only exponentials — solve the equations we meet at the end of today. It is worth over-learning.

■ Watch out

$e^{-t/2}$ is *not* a power of t , so the power rule has nothing to say about it. Its derivative is $-\frac{1}{2}e^{-t/2}$.

Answers shaped like $(-t/2)e^{(-t/2)-1}$ come from applying $t^n \rightarrow nt^{n-1}$ to an exponential — treating the exponent as if it were the power.

The difference in one question: **where is the variable — in the base, or in the exponent?** In t^2 it is the base, so the power rule applies. In e^{2t} it is the exponent, so the boxed rule applies.

$f' > 0 \Rightarrow$ rising, $f' < 0 \Rightarrow$ falling, $f' = 0 \Rightarrow$ momentarily flat

— a peak, a trough, or a value that has stopped moving altogether.

■ Intuition

To find where a function is largest: set $f' = 0$, solve, and check the sign of f' on either side. Positive then negative is a maximum; negative then positive is a minimum.

Very often you never need to evaluate anything — the sign answers the question.

■ Example

Differentiate $f(t) = 3t^2 + 2t + 5$, then $R(t) = 0.9e^{-t/2}$, then $P(t) = 100 - 80e^{-t/10}$.

Polynomial, term by term: $3t^2 \rightarrow 6t$, $2t \rightarrow 2$, $5 \rightarrow 0$, so $f'(t) = 6t + 2$. At $t = 0$ the slope is 2; at $t = -\frac{1}{3}$ it is zero, the bottom of the parabola.

Exponential: the number in front of t in the exponent is $a = -\frac{1}{2}$, so $R'(t) = -\frac{1}{2} \times 0.9e^{-t/2} = -\frac{1}{2}R(t)$. Read that: *the rate of forgetting is proportional to how much you still remember*.

The sum: the constant 100 gives 0, and the -80 comes out in front, so $P'(t) = +8e^{-t/10}$ — positive throughout, and largest at $t = 0$. Performance keeps improving, fastest at the start.

■ Practice — try it

1. Differentiate $f(t) = t^2 - 4t + 7$. Where is its slope zero, and is that a peak or a trough?
2. Differentiate $g(t) = e^{3t}$ and $h(t) = 5e^{-2t}$. Which grows and which decays?
3. Differentiate $R(t) = 0.8e^{-t/4}$ and check that $R' = -R/4$.
4. Differentiate $P(t) = 100 - 60e^{-t/5}$. Is P rising or falling?

■ Definition

$\int_a^b f(t) dt$ is the **area** between the graph of f and the horizontal axis, from $t = a$ to $t = b$ — counted positive above the axis and negative below it.

Two ways of reading it, both useful:

- chop the interval into thin slices of width dt , each contributing $f(t) \times dt$, and add them up. That is what the long S means: a sum.
- if f is a **rate**, the integral is the **total accumulated**. Speed integrates to distance; a rate of responding integrates to a number of responses.

■ Useful fact

If $F' = f$, then $\int_a^b f(t) dt = F(b) - F(a)$. Such an F is an **antiderivative** of f , and it is only fixed up to an added constant — hence the “ $+ C$ ” in $\int f(t) dt = F(t) + C$.

So the table of derivatives becomes a table of integrals, read right to left:

$$\int t^n dt = \frac{t^{n+1}}{n+1} \quad (n \neq -1), \quad \int e^{at} dt = \frac{e^{at}}{a}, \quad \int \frac{dt}{t} = \ln |t|.$$

Always check an integral by differentiating your answer. It costs one line and it never fails to catch a slip.

■ Example

Compute $\int_0^2 (3t^2 - 4t + 1) dt$ and $\int_0^T e^{-t/\tau} dt$.

First: an antiderivative is $t^3 - 2t^2 + t$, so the integral is $(8 - 8 + 2) - (0) = 2$.

Second: an antiderivative of $e^{-t/\tau}$ is $-\tau e^{-t/\tau}$ (check by differentiating), so

$$\int_0^T e^{-t/\tau} dt = \tau(1 - e^{-T/\tau}) \xrightarrow{T \rightarrow \infty} \tau.$$

The total area under a decaying exponential is exactly its time constant. So τ is not only a clock: integrated over all time, the forgetting curve $R_0 e^{-t/\tau}$ delivers a total of $R_0 \tau$.

■ Practice — try it

1. Compute $\int_0^3 (2t + 1) dt$.
2. Compute $\int_0^\infty e^{-2t} dt$. What does the answer equal, in words?
3. Find an antiderivative of $e^{-t/5}$, and check it by differentiating.

■ Definition

A function of **several variables** takes a list of numbers and returns one number, for example $f(x, y)$. Its graph is a **surface**: the height $z = f(x, y)$ above each point of the plane.

You already meet these constantly:

- accuracy as a function of task difficulty *and* amount of practice;
- a total score built from three subscales;
- how badly a model fits, as a function of its parameters.

With three inputs or more you can no longer draw the surface — and you will not need to. Everything we do works the same way regardless.

■ Definition

A **contour line** (or level set) is the set of points where f takes one fixed value: all the (x, y) with $f(x, y) = c$. Drawing several of them, for a ladder of values, gives the **contour map**.

Read it exactly like a walking map:

- contours *close together* $\Rightarrow$ a steep slope;
- contours *far apart* $\Rightarrow$ a flat region;
- a closed loop shrinking to a point $\Rightarrow$ a summit or a basin.

■ Link to cognitive science

An *iso-performance* curve — all the combinations of difficulty and practice that give the same accuracy — is exactly a contour line.

■ Definition

The **partial derivative** $\frac{\partial f}{\partial x}$ measures how f changes when you nudge *only* x , holding the other variables fixed. So you differentiate with the ordinary rules, treating the other letters as constants.

■ Example

$$f(x, y) = x^2 + 3xy + y. \quad \frac{\partial f}{\partial x} = 2x + 3y, \quad \frac{\partial f}{\partial y} = 3x + 1.$$

The recipe, said crudely and correctly: **cover the other letters with your thumb and differentiate**. The curly ∂ is not a new operation — it is only a reminder that other variables exist.

■ Definition

A **vector-valued function** of one variable returns several numbers at once:

$$\mathbf{x}(t) = (x(t), y(t)).$$

You differentiate it **one component at a time**: $\dot{\mathbf{x}}(t) = (\dot{x}(t), \dot{y}(t))$, and the result is again a vector.

As t runs, $\mathbf{x}(t)$ traces a **path**, and $\dot{\mathbf{x}}(t)$ is the **velocity**: it points along the path in the direction of travel, and its length is the speed.

■ Link to cognitive science

In a mouse-tracking experiment the cursor position $(x(t), y(t))$ is recorded every few milliseconds as the participant moves toward one of two response buttons. A path that bends toward the *unchosen* option before committing is standard evidence that the two responses competed — and “bends toward” is a statement about the velocity.

■ Practice — try it

1. For $f(x, y) = x^2 + 3xy + y$, compute both partial derivatives, then evaluate them at $(1, 2)$.
2. For $f(x, y) = x^2 + 4y^2$, compute both partial derivatives. Sketch the contour line $f = 4$.
3. For $\mathbf{x}(t) = (2t, 3 - t)$, compute $\dot{\mathbf{x}}$ and the speed. What shape is the path?
4. For $\mathbf{x}(t) = (t, e^{-t})$, compute $\dot{\mathbf{x}}$ and give the velocity at $t = 0$.

*We are told how fast a quantity is changing, in terms of the quantity itself.
We want to know what the quantity does.*

A rule of that kind is a **differential equation**, and to **solve** it means to produce a function $x(t)$ which, when differentiated, gives back exactly what the equation says.

■ Watch out

Faced with $\dot{x} = -(x - 3)$ it is tempting to anti-differentiate the right-hand side and write $-x^2/2 + 3x$. That answers a different question. Integration applies when the rate depends on *time*, $\dot{x} = f(t)$. Here the rate depends on x *itself*: the unknown is on both sides, and there is nothing to integrate yet.

■ Definition

$$\frac{dx}{dt} = a x \quad \implies \quad x(t) = x_0 e^{at}, \quad x_0 = x(0).$$

Check it with the boxed rule: differentiating $x_0 e^{at}$ gives $a x_0 e^{at}$, which is $a x$. ✓

The equation asked for a function proportional to its own derivative, and the exponential is the only shape that does that. The sign of a decides the fate: $a < 0$ decay to zero, $a > 0$ runaway growth.

We met this equation already: $R' = -R/\tau$, the forgetting curve. It was a differential equation all along.

■ Definition

$$\frac{dx}{dt} = ax + b \quad \implies \quad x(t) = x^* + (x_0 - x^*)e^{at}, \quad x^* = -\frac{b}{a} \quad (a \neq 0).$$

Read the answer as a sentence: *start at x_0 , end at x^* , and close the gap exponentially.*

- x^* is the **steady state**: set the rate to zero and solve. There the quantity stops moving.
- $a < 0$: the gap shrinks — **stable**, the system returns if disturbed.
 $a > 0$: the gap grows — **unstable**.
- $\tau = -1/a$ is the **time constant**, as before.

■ Link to cognitive science

Performance P improves with practice, but not without limit: there is a ceiling $P_{\max}$, and the closer you get, the slower you improve. As an equation,

$$\frac{dP}{dt} = k(P_{\max} - P),$$

which is $\dot{x} = ax + b$ with $a = -k$, so $P^* = P_{\max}$ and

$$P(t) = P_{\max} - (P_{\max} - P_0)e^{-kt}.$$

With $P_0 = 20$, $P_{\max} = 100$, $k = 0.1$ **per session**: the time constant is $\tau = 1/k = 10$ sessions. After 10 sessions, $P = 100 - 80 \times 0.37 \approx 70$; after 30, $P \approx 96$.

This is the forgetting curve turned upside down — same equation, same exponential, opposite direction. Both say: *the rate is proportional to the distance still to go*.

■ Practice — try it

1. Solve $\dot{x} = -2x$ with $x_0 = 5$. Where does x go as t grows?
2. For $\dot{x} = -x + 3$: find x^* and τ , then write $x(t)$ starting from $x_0 = 0$. Is it stable?
3. Check by substitution that $x(t) = 3 + C e^{-t}$ solves $\dot{x} = -(x - 3)$ for every constant C .
4. Learning obeys $\dot{P} = 0.2(90 - P)$ with $P_0 = 30$. Give τ , write $P(t)$, and compute $P(5)$.

- A **function** has a variable and some parameters; the **exponential** e^{at} is the shape that matters, and τ is its clock — 37% left after one, about 5% after three.
- The **derivative** is a slope; its sign says rising, falling or stopped; and $e^{at} \rightarrow a e^{at}$ is the rule everything rests on.
- The **integral** is an area, and the total accumulated from a rate. It is differentiation run backwards.
- With **several variables**: contour maps and *partial derivatives* — differentiate one letter at a time. With a **vector output**: a path, and its velocity, one component at a time.
- $\dot{x} = ax + b$ starts at x_0 , ends at $x^* = -b/a$, and closes the gap exponentially. *Solving* it is not *integrating* it, and $a < 0$ means stable.

Forgetting and learning are the same equation. If you keep one thing from today, keep that.